Manohar

Researcher

MANOHAR KUMAR C. V. S. S.
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Full name: Venkata Shiva Satya Manohar Kumar Changalagari

As a *researcher*, I've worked with a wide range of *remote sensing datasets*, including multispectral and hyperspectral imagery, as well as LiDAR point cloud, SAR, and thermal imagery. I intended to investigate *integrating datasets from different kinds of remote sensing sensors* for applications in a wide range of sectors, including forestry, agriculture, target detection and identification, and environmental monitoring. I sincerely appreciate *natural landscapes*, so I pursued a master's degree in remote sensing. In my spare time, I like exploring *new places, photographing natural landscapes and live moments, and making new friends.*

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Research Interests

Focus: **AI learning models for Agriculture, Forestry, Water Resources, Target Detection and Estimation, Environmental Biodiversity and Pollutants (plastic to microplastics)**

Data: **Earth Observation Remote Sensing data (i.e., Multispectral, Hyperspectral, LiDAR, and SAR)**

Platforms: **Terrestrial, Drone, Airborne, and Spaceborne**

Field: **Good field experience in the designing, planning and data acquisition**

Integration of different types of geospatial remote sensing data

Teaching and Research Experience

Vellore Institute of Technology, Vellore, India

2025 – ***Department of Quantum AI***, *School of Computer Science and Engineering*

Teaching and mentoring the bachelor and master students in their research

Central University of Andhra Pradesh, Ananthapuramu, India

2024 – 2025 ***Department of Space Science and Technology***

Teaching and mentoring the master students in their research

Indian Institute of Space Science and Technology, Thiruvananthapuram, India

2021 – 2024 ***Spectral Unmixing and its Applications (Agriculture, Forestry, Target Detection, and Estimation of Microplastics in Beach environment)***

Advisor : **Dr. Rama Rao Nidamanuri**, *Professor, Department of Earth and Space Sciences*

Indian Institute of Space Science and Technology, Thiruvananthapuram, India

2016 – 2021 ***Development of a stand-alone atmospheric correction module for hyperspectral data for the Indian context***

Advisor : **Dr. Rama Rao Nidamanuri**, *Professor, Department of Earth and Space Sciences*
Indian Institute of Technology, Gandhinagar, India

2015 – 2016 ***Sediment dynamics and sediment connectivity in the Kosi basin: Implications for river hazards***

Advisor : **Prof. Vikrant Jain** , *Discipline of Earth Sciences*
Indian Institue of Remote Sensing, Dehradun, India

2014 – 2015 ***Developing a framework for water feature extraction at variable resolution***

Advisor : **Dr. Poonam Seth Tiwari & Dr. Hina Pande** , *Scientist - SF, Photogrammetry and Remote Sensing Department*

Education

2024 : **Ph.D., Remote Sensing and Image Processing**, *Indian Institute of Space Science and Technology*, Thiruvananthapuram, India, (Supervisor: Professor Rama Rao Nidamanuri)
Dissertation: Benchmark Studies on Spectral Unmixing of Multi-sensor Hyperspectral Imagery

2015 : **Master of Technology, Remote Sensing & Geographical Information System with Specialization in Satellite Image Analysis and Photogrammetry, 79.7%**, *Indian Institute of Remote Sensing*, Dehradun, India
Dissertation: Automatic Extraction of Water Features at Multiple Scales

2011 : **Bachelor of Technology, Electronics & Communication Engineering, 75.25%**, *Jawaharlal Nehru Technological University Anantapur*, India

2006 : **Higher Secondary Education, 85.4%**, *Board of Intermediate Education, Andhra Pradesh*, India

2004 : **Secondary Education, 84%**, *Board of Secondary Education, Andhra Pradesh*, India

Publications

Journal Articles

2026 **H. Chaitra, Sivaganesh, C.V.S.S. Manohar Kumar, R.G. Sharathchandra, Priya Narayanan, and Rama Rao Nidamanuri**, Machine Learning-Based Detection of Fusarium Wilt in Tomato using Ground-based and Satellite Hyperspectral Imagery, ***Advances in Space Research,*** In Press, Journal Pre-proof, Elsevier, 2026.

2024 **C.V.S.S. Manohar Kumar, Sudhanshu Shekhar Jha, Rama Rao Nidamanuri, and Vinay Kumar Dadhwal**, Precision crop mapping: within plant canopy discrimination of crop and soil using multi-sensor hyperspectral imagery, ***Scientific Reports,*** Volume 14, pages 24903, Nature Portfolio, 2024.

2024 **Vamshi Krishna Munipalle , Usha Rani Nelakuditi, Manohar Kumar CVSS, Rama Rao Nidamanuri**, Ultra-high-resolution hyperspectral imagery datasets for precision agriculture applications, ***Data in Brief,*** Volume 55, page 110649, Elsevier, 2024.

2023 **C.B. Rajesh , C. V. S. S. Manohar Kumar, Sudhanshu Shekhar Jha, K.I. Ramachandran, and Rama Rao Nidamanuri**, In-situ and airborne hyperspectral data for detecting agricultural activities in a dense forest landscape, ***Data in Brief,*** Volume 50, page 109510, Elsevier, 2023.

2022 **C.V.S.S. Manohar Kumar, Sudhanshu Shekhar Jha, Rama Rao Nidamanuri, and Vinay Kumar Dadhwal**, Benchmark studies on pixel-level spectral unmixing of multi-resolution hyperspectral imagery, ***International Journal of Remote Sensing,*** Volume 43, pages 1451-1484, Taylor and Francis Ltd., 2022.

2022 **CVSS Manohar Kumar , Sudhanshu Shekhar Jha, and Rama Rao Nidamanuri**, Multi-resolution terrestrial hyperspectral dataset for spectral unmixing problems, ***Data in Brief,*** Volume 43, page 108331, Elsevier, 2022.

2020 **Sudhanshu Shekhar Jha , Manohar Kumar , and Rama Rao Nidamanuri**, Multi-platform optical remote sensing dataset for target detection, ***Data in Brief,*** Volume 33, page 106362, Elsevier, 2020.

2019 **Sudhanshu Shekhar Jha, C.V.S.S. Manohar Kumar, and Rama Rao Nidamanuri**, Flexible atmospheric compensation technique (fact): a 6s based atmospheric correction scheme for remote sensing data., ***Geocarto International,*** Volume 36, pages 28-46, Taylor and Francis Ltd., 2019.

Conference Proceedings

2026 **Suresh Nagarajan S, Manohar Kumar C V S S** , Plant-Level Segmentation Using Mask R-CNN with Transformer-Based Feature Enhancement for Multi-Resolution Drone Hyperspectral Imagery, In ***4th International Conference on Machine Intelligence for GeoAnalytics and Remote Sensing,*** MIGARS 2026, Thiruvananthapuram, Kerala, India, IEEE GRSS, 2026.

2026 **Rakshana B, Suvathi S, Sowmiya K, Manohar Kumar C V S S**, Zone-Adaptive Stacking Ensemble with Spectral-Texture Feature Fusion for Chlorophyll-A Retrieval in Chilika Lake Using Sentinel-2 Imagery, In ***4th International Conference on Machine Intelligence for GeoAnalytics and Remote Sensing,*** MIGARS 2026, Thiruvananthapuram, Kerala, India, IEEE GRSS, 2026.

2023 **C.V.S.S. Manohar Kumar, Rama Rao Nidamanuri, and Vinay Kumar Dadhwal**, Sub-pixel discrimination of soil and crop in drone-based hyperspectral imagery, In ***13th workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing,*** whispers 2023, Athens, Greece, IEEE GRSS, 2023.

2023 **C. V. S. S. Manohar Kumar, M. S. Salini, and Rama Rao Nidamanuri.**, Abundance of plastic-litter in hyperspectral imagery using spectral unmixing in coastal environment, In ***IEEE International Geoscience and Remote Sensing Symposium,*** IGARSS 2023, Pasadena, California, USA, IEEE GRSS, 2023.

2023 **C. V. S. S. Manohar Kumar, Sudhanshu Shekhar Jha, and Rama Rao Nidamanuri**, Target detection in airborne hyperspectral imagery and its sensitivity to different atmospheric correction methods., In ***International Conference on Machine Intelligence for GeoAnalytics and Remote Sensing,*** MIGARS 2023, Sponsered by IEEE GRSS, ACM Hyderabad, at Vasavi College of Engineering, Hyderabad, Telangana, India, IEEE GRSS, 2023.

2023 **C. V. S. S. Manohar Kumar, Rama Rao Nidamanuri, and Vinay Kumar**, Subpixel level discrimination of vegetable crops in a complex landscape environment, In ***International Conference on Machine Intelligence for GeoAnalytics and Remote Sensing,*** MIGARS 2023, Sponsered by IEEE GRSS, ACM Hyderabad, at Vasavi College of Engineering, Hyderabad, Telangana, India, IEEE GRSS 2023.

2023 **P Nijitha, C. V. S. S. Manohar Kumar, S Pankaj Dhanya, and Nidamanuri Rama Rao**, Solid waste detection and waste-material characterization in urban environment at subpixel level in airborne hyperspectral imagery, In ***13th workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing,*** whispers 2023, Athens, Greece, IEEE GRSS, 2023.

2023 **K. C. Indu, C. V. S. S. Manohar Kumar, S Pankaj Dhanya, and Nidamanuri Rama Rao**, Automatic object-based plant-level crop segmentation in drone-based hyperspectral imagery, In ***13th workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing,*** whispers 2023, Athens, Greece, IEEE GRSS, 2023.

2023 **B. R. Aarsha, C. V. S. S. Manohar Kumar, S Pankaj Dhanya, and Nidamanuri Rama Rao**, Invasive plant species detection in airborne hyperspectral imagery over complex forest landscape, In ***13th workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing,*** whispers 2023, Athens, Greece, IEEE GRSS, 2023.

2023 **Feba Treasa Jose, C. V. S. S. Manohar Kumar, and Rama Rao Nidamanuri**, Influence of atmospheric correction models on the discrimination of crops using airborne hyperspectral imagery, In ***International Conference on Machine Intelligence for GeoAnalytics and Remote Sensing,*** MIGARS 2023, Sponsered by IEEE GRSS, ACM Hyderabad, at Vasavi College of Engineering, Hyderabad, Telangana, India, IEEE GRSS, 2023.

2023 **Sivaganesh, Chaitra H, Manohar Kumar C. V. S. S., Manoj Kaushik, R. G. Sharathchandra, and Rama Rao Nidamanuri**, Hyperspectral detection of fusarium wilt in tomato plants using machine learning-based approaches, In ***IEEE International Geoscience and Remote Sensing Symposium,*** IGARSS 2023, Pasadena, California, USA, IEEE GRSS, 2023.

Conference Presented

2020 **C. V. S. S. Manohar Kumar, Rama Rao Nidamanuri, and Vinay Kumar Dadhwal**, Dynamics of spectral abundances with respect to change in sources of end-members, In ***IEEE India Geoscience and Remote Sensing Young Researchers Conclave 2020,*** Indian Institute of Space Science and Technology, Thiruvananthapuram, Kerala, India, 2020.

2018 **C. V. S. S. Manohar Kumar, Rama Rao Nidamanuri, and Vinay Kumar Dadhwal**, Atmospheric water vapour estimation for surface reflectance retrieval from AVIRIS-NG hyperspectral imagery, In ***Impact of Climate Change on Water Resource,*** ICWR'2018, Periyar University, Salem, Tamilnadu, India, 2018.

Workshops & Schools

Indian Institute of Technology Madras, Chennai, India

2025 ***Completed the "Winter School on Geospatial Technologies (Level 2: Advanced Geospatial Techniques for Coastal and Marine Applications" sponsored by Department of Science and Technology (DST) - Natural Resources Data Management System (NRDMS), Government of India, New Delhi.***

Space Application Center, Ahmedabad, India

2023 ***Participated in "NISAR Science Workshop" organized by Space Application Center, Department of Space, Government of India, New Delhi.***

Indian Institue of Space Science and Technology, Thiruvananthapuram, India

2023 ***Completed the "Winter School on Geospatial Science and Technology (Level 2: Theme - Water and Environment Pollution Monitoring using Multi-source Remote Sensing" sponsored by National Geospatial Program - Department of Science and Technology, Government of India, New Delhi.***

2019 ***Completed the "Winter School on Orientation to Geospatial Technologies: GEOSMART 2019" sponsored by Department of Science and Technology (DST) - Natural Resources Data Management System (NRDMS), Government of India, New Delhi.***

Indian Institue of Remote Sensing, Dehradun, India

2014 ***Completed the advanced Course on "Microwave (SAR) Remote Sensing for Natural Resources"***

Andhra University, Visakhapatnam, India

2012 ***Participated a National level workshop on "Recent Advances in Statistical Signal Processing and its Applications"***

Shree Institute of Technical Education, Tirupathi, India

2011 ***Participated in 2 week "Faculty Development Programme in Entrepreneurship"***

International Institute of Information Technology, Hyderabad, India

2010 ***Participated in 2nd National Workshop for Students on "Usability and User Interface Design", TechEase'10***

Fellowships & Funds

2021 – 2024 ***Senior Research Fellowship - Direct***, Council of Scientific and Industrial Research (CSIR), Government of India

2018 – 2021 ***Senior Research Fellowship***, by Department of Science and Technology (DST), Government of India

2016 – 2018 ***Junior Research Fellowship***, by Department of Science and Technology (DST), Government of India

2014 – 2015 Recipient of ***ISRO, EOAM***, project fund by Department of Space (DoS), Government of India

2013 – 2015 Recipient of ***GATE Fellowship*** by Ministry of Human Resource Development (MHRD), Government of India

Grants Received

2023 ***International Travel Grant***, by IEEE Geoscience and Remote Sensing Society (GRSS)

2023 ***International Travel Grant***, by Department of Science and Technology, Science and Engineering Research Board (DST - SERB), Government of India

2023 ***CSIR Foreign Travel Grant***, by Council of Scientific and Industrial Research, Human Resource Development Group (CSIR - HRDG), Government of India

Awards & Recognitions

2023 **The best poster award** at ***International Conference on Machine Intelligence for GeoAnalytics and Remote Sensing (MIGARS) 2023 by IEEE Geoscience and Remote Sensing Society at Vasavi College of Engineering,*** Hyderabad, India

2010 **The best presentation award** at ***41st Mid-term Symposium of National level student paper contest conducted by IETE at Sri Venkateswara University,*** Tirupati, India

Position of Responsibility

2020: **Instructor for the IEEE GRSS course on Programming Approach in Remote Sensing**, IEEE GRSS Kerala Chapter

2017 - 2019: **Organizing and taking sessions in the yearly event "GeoConnect"**, Indian Institute of Space Science and Technology, Thiruvananthapuram, India

Teaching Assistantship

Indian Institute of Space Science and Technology, Thiruvananthapuram, India

Fall, 2017-23: **ESG631: Remote Sensing and Image Analysis**

Spr, 2017-23: **ESG623: Microwave Remote Sensing Data Processing**

Fall, 2019: **ESG616: Scientific Computing for Geospatial Data Analysis**

Technical skills

Programming: Python, MATLAB, R

Cloud: GoogleEarthEngine, MicrosoftPlanetaryComputer

Software: SNAP, ENVI, ArcGIS, Erdas Imagine, QGIS, Cloud Compare, Terrasolid.

Platforms: Windows, Linux, Mac OS.

Instruments: Hyperspectral Imager (Headwall VNIR-A, Cubert FireFly S 185), DGPS (Trimble R10, and R12), Terrestrial Laser Scanner (FARO Focus S 350, RIEGL VZ-400), Spectro-radiometers (ASD and SVC).

Activities

Sports: **Badminton, Running, Cycling**

Personal: **Photography and Travelling**

Languages

Education: **English**, *Speak, Read and Write*

Mother: **Telugu**, *Speak, Read and Write*

Official: **Hindi**, *Speak, Read and Write*

References

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Indian Institute of Space Science and Technology (IIST)
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National Remote Sensing Centre (NRSC),
Indian Space Research Organisation (ISRO)
Hyderabad, India

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Mr. Vinay Kumar

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Indian Institute of Remote Sensing (IIRS),
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